Duygu Genez

000908732

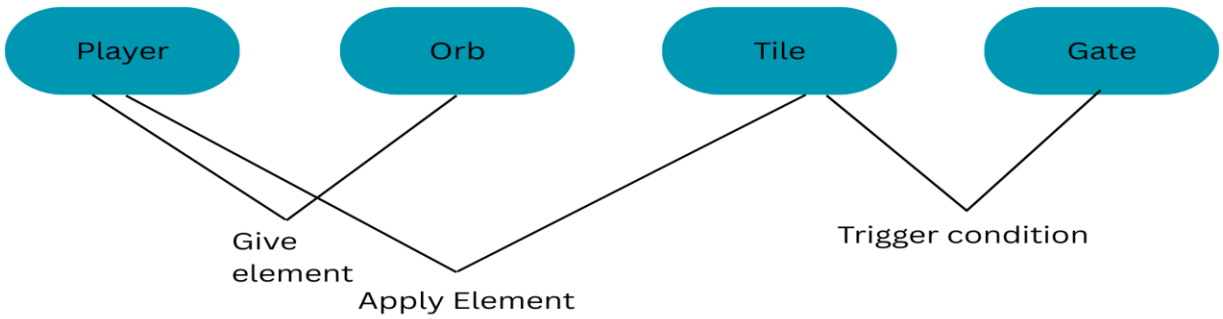
Systemic Games with Emergent Mechanics

Echoes of the Elemental Vault

An element system based on very few types of objects, designed to create emergent puzzle interactions as a result of the contextual properties within that object type.

Object	*Properties*	*Behaviours*	*Relationship*
Player	• Health 0-5 – held element (fire,shock,water) – Stamina 0-100	Element/pulse/push/attack	Gets element from orb / changes tile state and triggers gate indirectly
Elemental orb	• Element type (fire,water,shock) – state (charged) – Conductive (true, false)	transfer element/react to elements	gives power to players/affected by tile state
environmental Tile	• Surface (water, grass,metal,stone) – burn state (unburned/burning)	conduct shock/ spread fire	controls elemet speard / triggers gate condition
20	• Activation(fire,shock) – State (closed/open)	block path open when condition met	Responds to tile state

Input	*System Check*	*State Change*	*Output*	*Feedback*
Press element pulse	is tile wet or burned?	shock radius increases	gate conditions met	blue flash - sound -gate opens



Echoes of the Elemental Vault *created with Canva*

System Summary:

- The system consists of a limited amount of object.
- Interactions are determined by the context in which they occur.
- The gate will depend upon the state of the environment rather than directly on the player.
- Feedback will aide in the creation of a mental model by the player.

Scenario #1 – Shock Puzzle

Arrangements:

- Character, electric orb, metal tiles, wet tiles, gateway (distant)

Description of Events:

-Character gets electric orb to use

-Character releases a shockwave effect

-Electric wave travels through metal

-As wet tile is added, shock expands outward

-As gate opens, shock will cause it to open

Emerging Features:

Using shock alone does not create an effect and shock and wet are required to create an effect

Feedbacks:

- Electric shockwave blue trail
- Electric shockwave wet tile on
- Gate will be opened by blue pulse

Scenario #2 – Fire and Shock Puzzle

Arrangements:

- Tiles of water, tiles of wood, tiles of metal and gate

Actions:

- The player has caught on fire
- The water has evaporated
- There is an underlying layer of metal to be found.
- The player will receive an electric shock.
- Gate will be opened

Emergent Property:

Two Mechanical Combination Solutions are required and no single mechanical solution will work.

Design Rationale:

The system was intentionally built around the idea of having very few, or four core, object types present in it so that instead of having a very high number of mechanics create complexity, a high

level of contextual properties associated with each tile (i.e. what is happening around that tile) provides depth to the system. In this sense, each tile serves as the system mediator between player actions and the resulting environmental outcome. As such, gates do not respond to player inputs directly. Rather they respond to the changes that occur in the environment and thus reinforce indirect causality and a systemic behavior.

Consistent with this idea of using elemental properties (e.g. fire and shock) along with environmental states (e.g. wet, conductive, burning) provides multiple emergent puzzle combinations that yield greater than individual mechanic combinations.

The feedback provided to the player (such as color changes, light pulses, sound cues, and UI element indicators) collectively provides the player the ability to build a solid mental model of the cause and effect relationships within the system.